Anish Parepalli

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Education

University of North Carolina at Chapel Hill — Charlotte, NC
B.S. in Computer Science, Statistics — *August 2022 – May 2026*

- **GPA**: 3.60 **Teaching Assistant:** Foundations of Programming (Java)
- **Coursework:** Data Structures & Algorithms, Artificial Intelligence, Software Engineering, Databases, Probability, Linear Algebra

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, SQL, Swift, C/C++
Frameworks: React, Next.js, FastAPI, Flask, Node.js, Angular, Pandas, PyTorch
AI & Cloud: OpenAI API, Anthropic Claude API, AWS Lambda, AWS Bedrock, DynamoDB, GCP, MCP
Databases & Tools: PostgreSQL, Supabase, REST APIs, Git, Jira, Docker, Kubernetes, CI/CD

Experience

Vanguard — May 2026 – Current
Application Engineer — *Charlotte, NC*

- Own end-to-end development of production features within Advice and Wealth Management Technology, from design and implementation through deployment, maintenance, and production support.
- Replaced a **Netflix Falcor** dependency with direct in-house API integration using **TypeScript**, improving latency **2–3x** across **900K+ monthly API calls** in a large-scale production system.
- Designed and shipped a **Claude-powered AI productivity tool** integrating **Atlassian MCP** to automate release checklists, code-quality summaries, and rollback plans, eliminating **5–10 minutes** of manual work per near-daily deployment.
- Built the AI execution layer for **tool invocation, structured output parsing, and error retries**, creating a reusable pattern for AI-assisted workflows later adopted by other engineering teams.
- Presented the product and development workflow to **15+ cross-functional and leadership stakeholders**, gathering feedback and driving team-wide adoption.

Vanguard — May 2025 – Aug 2025
Application Developer Intern — *Charlotte, NC*

- Developed **serverless distributed workflows** using **TypeScript and AWS Lambda** in an Agile engineering team, improving the accuracy and reliability of daily portfolio-management workflows supporting **$10T+ AUM**.
- Built a detection workflow that identified unenrolled clients and automatically corrected downstream reporting, improving daily reports used by **200+ advisors**, senior technology leaders, and executives.

Computer Science + Social Good — Aug 2023 – Aug 2024
Software Engineer Intern — *Chapel Hill, NC*

- Built and maintained backend services with **FastAPI and PostgreSQL**, optimizing database queries and API logic to reduce response time by **40%** for a large-scale ticketing platform.
- Automated onboarding and access provisioning for **200+ users** using Firebase, reducing setup time by **60%** and improving consistency of user access across research teams.

Projects

AI-Powered FAQ System for UNC Courses | *Python, Angular, FastAPI, OpenAI API*

- Built a **retrieval-based AI assistant** using the **OpenAI API** to help students answer course questions, reducing duplicate office-hour questions by **35%** across 20 computer science courses.
- Developed the **Python/FastAPI backend** and web frontend, implementing REST APIs, authentication, role-based access control, and LLM-powered retrieval workflows.

Oriole | *Next.js, React, TypeScript, Supabase, PostgreSQL, Drizzle, tRPC*

- Built and shipped a full-stack web application used by **200+ users**, developing customer-facing features across the **React/Next.js frontend, TypeScript backend, and PostgreSQL database**.
- Designed type-safe APIs with **tRPC and Zod** and implemented database operations with **Drizzle/Supabase**, owning features from UI design through backend implementation and deployment.

Tax Optimization Web App | *Python, Flask, AWS Bedrock, DynamoDB, Claude*

- Built a financial web application that automated state tax-exempt dividend analysis across **300+ mutual funds**, eliminating **100%** of manual lookup work.
- Developed a **Python REST API** with AWS DynamoDB supporting concurrent requests with **sub-200ms latency** and integrated Claude through AWS Bedrock to generate automated tax-planning summaries.
- Automated financial data extraction and transformation workflows, reducing tax-planning research time by **50%**.